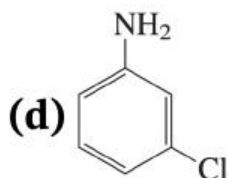
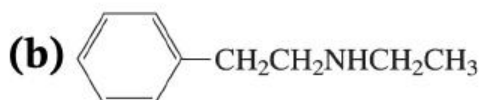
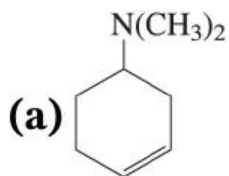


Solutions to Problems

27.

- (a) 3-Hexanamine, 3-aminohexane;
- (b) *N*-methyl-2-propanamine, 2-(methylamino)propane, isopropylmethylamine;
- (c) 2-chloroaniline, *o*-chloroaniline;
- (d) *N*-methyl-*N*-propylbenzenamine; *N*-methyl-*N*-propylaniline;
- (e) *N,N*-dimethylmethanamine (common: trimethylamine); *N,N*-dimethylaminomethane;
- (f) 4-(*N,N*-dimethylamino)-2-butanone (only satisfactory name);
- (g) 6-chloro-*N*-cyclopentyl-*N*,5-dimethyl-1-hexanamine (numbers refer to substituents on parent hexane chain); 1-chloro-6-(*N*-cyclopentyl-*N*-methylamino)-2-methylhexane;
- (h) *N,N*-diethyl-2-propen-1-amine, 3-(*N,N*-diethylamino)-1-propene

28.



29.

(a) 1-Cyclohexyl-N-methyl-2-propanamine;

(b) 1-phenyl-2-propanamine;

(c) 2-(3,4,5-trimethoxyphenyl)ethanamine;

(d) *R*-4-(1-hydroxy-2-[*N*-methylamino]ethyl)benzene-1,2-diol.

30.

(a) $5-7 \text{ kcal mol}^{-1}$, $5-7 \text{ kcal mol}^{-1}$, approximately equal to E_a for inversion.

(b) Methyl anion is isoelectronic with ammonia and, likewise, is tetrahedral (sp^3 hybridized). Methyl radical and cation, with one and two fewer electrons, respectively, are more stable when trigonal planar. They gain bond strength by rehybridizing to use sp^2 orbitals in the σ bonds, σ bonds, with either a singly occupied or vacant p orbital "left over." The sp^2 scheme is not as good for the anion or for the ammonia because two electrons in an unhybridized p orbital is quite unfavorable in the absence of other stabilizing influences: Such electrons are far from and attracted only poorly by the atom's nucleus.

31. The odd atomic weights suggest that each contains a single nitrogen.

The total number of hydrogens is available from the NMR, ¹H NMR, so the number of carbons can be determined by difference:

$m/z \ 129 = 14 \text{ (one N)} + 19 \text{ (19 H's)} + \text{weight of carbons.}$

$m/z \ 129 = 14 \text{ (one N)} + 19 \text{ (19 H's)} + \text{weight of carbons. Weight of}$

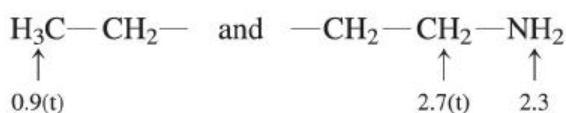
$\text{carbons} = 96 \rightarrow 8 \text{ carbons; Weight of carbons} = 96 \rightarrow 8 \text{ carbons; } C_8H_{19}N$

$C_8H_{19}N$ is the formula for each of these unknowns. Degree of

unsaturation (see [Section 11-8](#)): $H_{\text{sat}} = 16 + 2 + 1 \text{ (for the N)} = 19$;

$H_{\text{sat}} = 16 + 2 + 1 \text{ (for the N)} = 19$; compounds are saturated.

A ¹H NMR:



Notice that the signal at $\delta=2.7$ is **not** split by the $\text{—NH}_2\text{—NH}_2$ hydrogens (similar to the situation for alcohols).

The splittings nicely reveal the number of neighboring H's.

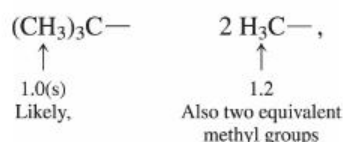
MS: m/z 30 for $[\text{H}_2\text{C}^+\text{—}\ddot{\text{N}}\text{H}_2 \leftrightarrow \text{H}_2\text{C}=\text{N}^+\text{H}_2]$

MS: m/z 30 for $[\text{H}_2\text{C}^+\text{—}\ddot{\text{N}}\text{H}_2 \leftrightarrow \text{H}_2\text{C}=\text{N}^+\text{H}_2]$ fragment. All that remains is to insert C_4H_8 , and the simplest way to do that is as

$\text{CH}_3(\text{CH}_2)_7\text{NH}_2$ (1-octanamine). Other isomers would show additional methyl signals in the NMR near $\delta=0.9\text{—}1.0$.

$\delta=0.9\text{—}1.0$.

B NMR:



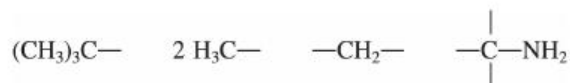
Perhaps a CH_2 and an NH_2 ? (Signals at $\delta=1.3$ and 1.4). MS: m/z 114 is $[\text{M—CH}_3]^+$, 72 is $[\text{M—}(\text{CH}_3)_3\text{C}]^+$,

MS: m/z 114 is $[\text{M—CH}_3]^+$, 72 is $[\text{M—}(\text{CH}_3)_3\text{C}]^+$, and 58 is most likely an iminium ion. Before guessing, notice that there are no NMR

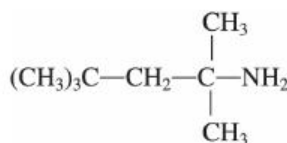
signals in the $\delta=2.7$ region, where you might expect to find $\text{—}\overset{\text{H}}{\underset{|}{\text{C}}}\text{—N}$

signals. So, most likely the N is attached to a **tertiary** carbon.

Possible pieces:



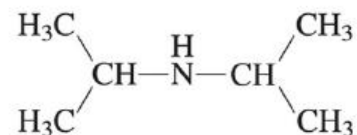
All atoms in the formula are present, so put it together



The m/z 58^{m/z} 58 fragment therefore is $[(CH_3)_2C=NH_2]^+$.
 $[(CH_3)_2C=NH_2]^+$.

32. As you do each of these, keep the $C_6H_{15}N$ formula in mind.

(a) NMR:NMR: The $\delta=23.7$ peak may correspond to one or more than one *equivalent* H_3C-H_3C- groups, and the peak at $\delta=45.3$ is one or more equivalent $>CH-$ units (attached to N due to chemical shift). IR: A **secondary** amine, $-NH-$. No other signals are present, so attach as many of each as are necessary. The answer is



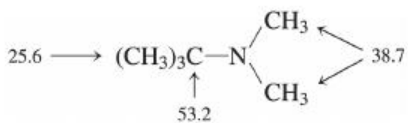
(b) NMR:NMR: Now you have only H_3C- and $-CH_2-$ groups (the latter attached to N); IR: A **tertiary** amine. So, the answer is $(CH_3CH_2)_3N$.
 $(CH_3CH_2)_3N$.

(c) NMR:NMR: H_3C-H_3C- groups, $-CH_2-CH_2-$ groups **not** attached to N, and $-CH_2-CH_2-$ groups that **are** attached to N. IR: Amine is **secondary**. So the answer is
 $CH_3CH_2CH_2NHCH_2CH_2CH_3$.
 $CH_3CH_2CH_2NHCH_2CH_2CH_3$.

(d) NMR:NMR: One H_3C-H_3C- and five $-CH_2-$'s.
 $-CH_2-$'s IR: **Primary** amine ($-NH_2$) ($-NH_2$). This is

—CH₂—s. IR: **Primary** amine (—NH₂). ν_{max} 3422. 1 ml is CH₃(CH₂)₅NH₂. CH₃(CH₂)₅NH₂.

(e) NMR:NMR: Two different $\text{H}_3\text{C}-\text{H}_3\text{C}-$ types, one ($\delta=38.7$) attached to N; also a quaternary CC attached to NN ($\delta=53.2$). IR: A **tertiary** amine. Remembering the $\text{C}_6\text{H}_{15}\text{N}$ formula, you can construct the molecule:

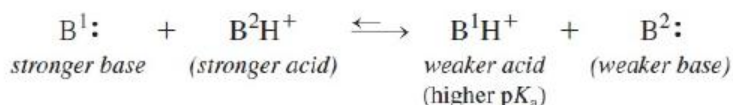


- 33.** Figure 21-5 is $(\text{CH}_3\text{CH}_2)_3\text{N}(\text{CH}_3\text{CH}_2)_3\text{N}$ for comparison purposes. Look in each case for important fragments from $\text{C}^+-\text{C}-\text{N}$ cleavage to make iminium ions.

(a) m/z 72^{m/z} 72 is important, which is $[M-29]^+$ or loss of $\text{CH}_3\text{CH}_2\text{—}$. The only amine that should easily lose an ethyl group from those in [Problem 28](#) is $\text{CH}_3\text{CH}_2\text{—CH}_2\text{—NH—CH}_2\text{CH}_2\text{CH}_3$ [see (c)]. This is the answer.

(b) m/z 86^{m/z} 86 is rather large, corresponding to the loss of $\text{H}_3\text{C—}$. Three amines in [Problem 32](#) should lose $\text{H}_3\text{C—}$ easily: (a), (b), and (e). *N,N*-Diethylethanamine [(triethylamine, b)] is ruled out because its MS (Figure 21-5) doesn't match. The m/z 58^{m/z} 58 peak is loss of 43, or C_3H_7 . That's easy to visualize from (a): $(\text{CH}_3)_2\text{CH—NH—CH}(\text{CH}_3)_2$ (the correct answer), but not from amine (e).

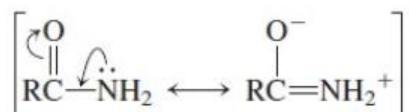
34. The base in question (call it “B1:” “B¹:”) is stronger. If its conjugate acid (B1H⁺) (B¹H⁺) has a high pK_a, pK_a , then that acid is weak; consequently, its corresponding conjugate base is strong. Here's the equation:



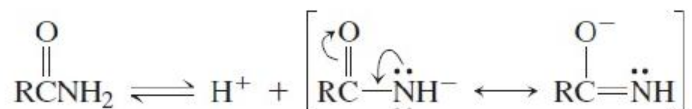
35. (a) To the left. NH_3 and $-\text{OH}$ are weaker acids and bases, respectively, than are H_2O and NH_2^- .
- (b) To the left. CH_3NH_2 is a weaker base than $-\text{OH}$, and H_2O is a weaker acid than CH_3NH_3^+ .
- (c) To the right. CH_3NH_2 is a stronger base (see textbook [Section 21-4](#)) than $(\text{CH}_3)_3\text{N}$.

36. (a)

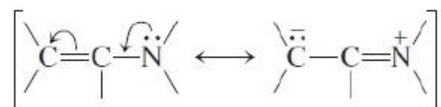
- (i) Weaker bases because lone pair on N is “tied up” by resonance:



Stronger acids because the conjugate base is stabilized by both the inductive effect of the carbonyl group and by resonance:



- (ii) Same as carboxamides, only to a greater extent for both acidity and basicity due to the two carbonyl groups.
- (iii) Somewhat weaker bases because of resonance:

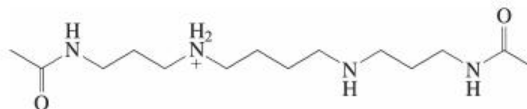


Not acidic, because of lack of H's on nitrogen.

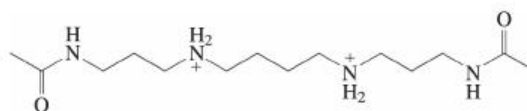
- (iv) Weaker bases and stronger acids, for the same reasons given in (a) for carboxamides.

(b) At $\text{pH} = 12$ the structure shown in the problem is the one present to the greatest extent.

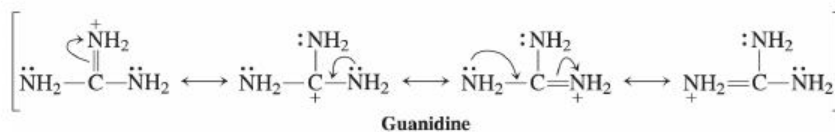
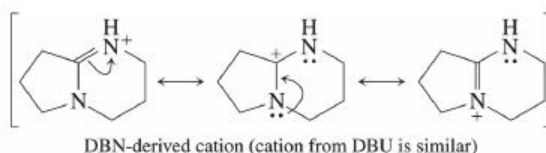
At $\text{pH} = 10.4$ the main form would have one of the amine nitrogens protonated:



At $\text{pH} = 7$ both amine nitrogens will be protonated:



37. Protonate a doubly bonded nitrogen in each case to get a resonance stabilized cation.



Resonance stabilization of conjugate acids enhances base strengths.

38.

(a) Excess NH_3 is the easiest way, also 1. N_3^- , 2. LiAlH_4 , or Gabriel (text, p. 1034);

(b) NH_3 , NaBH_3CN ; NH_3 , NaBH_3CN ;

(c) $\text{CH}_3\text{CH}_2\text{NH}_2$, NaBH_3CN ; $\text{CH}_3\text{CH}_2\text{NH}_2$, NaBH_3CN ;

(d) 1. $-\text{CN}$, 2. LiAlH_4 ;

(e) $(\text{CH}_3)_2\text{NH}$, NaBH_3CN ; $(\text{CH}_3)_2\text{NH}$, NaBH_3CN ;

(f) 1. $\text{H}_2\text{C}=\text{O}$, NH_3 , HCl , heat, 2. $-\text{OH}$, H_2O (Mannich);

(g) H_2 , PtO_2 , or LiAlH_4 ; H_2 , PtO_2 , or LiAlH_4 ;

(h) Br_2 , NaOH , H_2O , heat (Hofmann rearrangement).

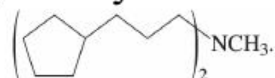
39.

(a) Not at all. This process **adds a carbon** (the $-\text{CN}$ group), making 1-pentanamine.

(b) Not at all. $\text{S}_\text{N}2$ reactions with tertiary haloalkanes are not possible.

(c) Well.

(d) Poorly. Further alkylation can occur, making



(e) Poorly. The haloalkane, although primary, is highly branched and will not react well in $\text{S}_{\text{N}}2$ reactions.

(f), (g) Well.

(h) Poorly. Four-membered rings are strained and difficult to form. The method would work well for a five- or six-membered ring.

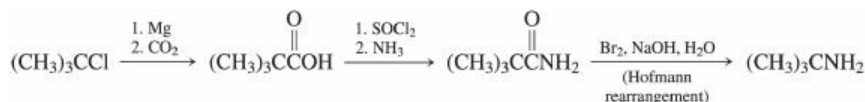
(i) Not at all. Reaction shown is for a benzene, not a cyclohexane compound.

(j) Well.

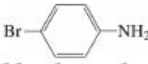
- 10.** $(\text{CH}_3\text{CH}_2)_2\text{O}$ solvent and H^+ , H_2O work-up are understood for Grignard and LiAlH_4 reactions.

(a) 1. NaN₃, DMSO, NaN₃, DMSO, 2. LiAlH₄. LiAlH₄.

(b) You need to be devious. Add a carbon and then take it out again!

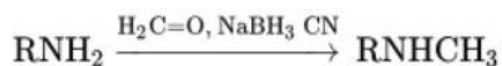


(c) 1. NaN₃, NaN₃, 2. LiAlH₄ LiAlH₄ (makes primary amine), 3. H₂C=O $\xrightarrow{\text{H}_2\text{C}=\text{O}}$ [makes imine; use one equivalent only, to avoid dimethylation ([Problem 56](#))], 4. NaBH₃CN, CH₃CH₂OH NaBH₃CN, CH₃CH₂OH (completes reductive amination). Note how CH₃CH₃ group is introduced as **formaldehyde**, with a subsequent reduction step.

(d) Same as (b). (h) No simple way to improve the situation. (i) Start with  and do reactions shown, making . The H₂, Pd/H₂, Pd ([Sections 14-7](#) and [15-2](#)) will slowly hydrogenate the ring.

11. CH₃CH₂NH₂, (CH₃CH₂)₂NH, (CH₃CH₂)₃N, and (CH₃CH₂)₄N⁺.
CH₃CH₂NH₂, (CH₃CH₂)₂NH, (CH₃CH₂)₃N, and (CH₃CH₂)₄N⁺.

12. Make pseudoephedrine from phenylpropanolamine using reductive amination. As suggested in [Problem 40\(d\)](#), use one equivalent of H₂C=O $\xrightarrow{\text{H}_2\text{C}=\text{O}}$ to avoid **dimethylation**.
RNH₂ → H₂C=O, NaBH₃CN RNHCH₃

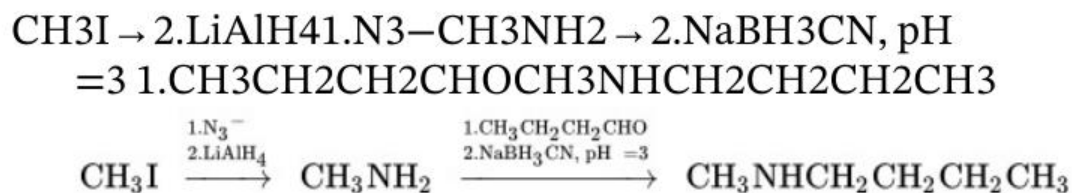


13. Secondary (general structure $\text{RR}'\text{NHRR}'\text{NH}$).

- (a)** 1. $\text{CH}_3\text{CH}_2\text{NH}_2$, H^+ , $\text{CH}_3\text{CH}_2\text{NH}_2$, H^+ , 2.
 NaBH_3CN , $\text{CH}_3\text{CH}_2\text{OH}$; NaBH_3CN , $\text{CH}_3\text{CH}_2\text{OH}$;
- (b)** 1. NaN_3 , DMF, NaN_3 , DMF, 2. LiAlH_4 , LiAlH_4 , THF, THF,
 3. CH_3CHO , H^+ , CH_3CHO , H^+ , 4.
 NaBH_3CN , $\text{CH}_3\text{CH}_2\text{OH}$; NaBH_3CN , $\text{CH}_3\text{CH}_2\text{OH}$;
- (c)** 1. SOCl_2 , SOCl_2 , 2. NH_3 , NH_3 , 3. Br_2 , NaOH , H_2O ,
 Br_2 , NaOH , H_2O , 4. CH_3CHO , H^+ , CH_3CHO , H^+ , 5.
 NaBH_3CN , $\text{CH}_3\text{CH}_2\text{OH}$; NaBH_3CN , $\text{CH}_3\text{CH}_2\text{OH}$

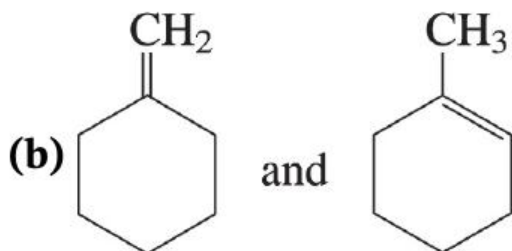
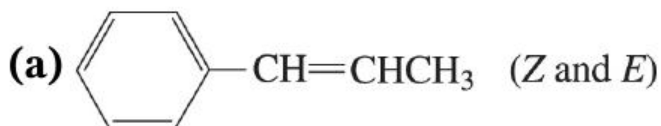
14.

- (a) Lots of ways! Bromobutane + azide, then LiAlH_4 reduction, or bromopropane + cyanide, then LiAlH_4 , LiAlH_4 , or bromobutane + phthalimide salt, then hydrolysis.
- (b) First make methanamine from iodomethane and either azide or phthalimide salt [similar to butanamine syntheses in (a)], and then carry out reductive amination using the methanamine and butanal:

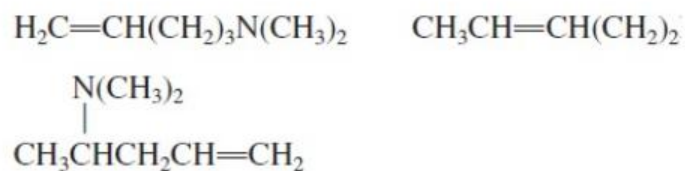


- (c) Make butanamine (a) and then go with double reductive amination using excess formaldehyde and NaBH_3CN . Best way to make any N,N -dimethyl N,N -dimethyl tertiary amines (see [Problem 56](#)).

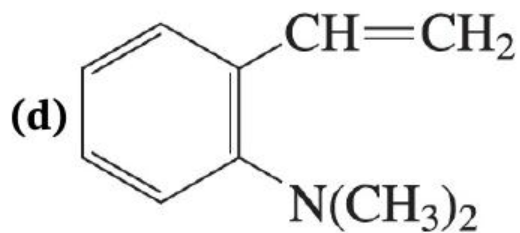
15.



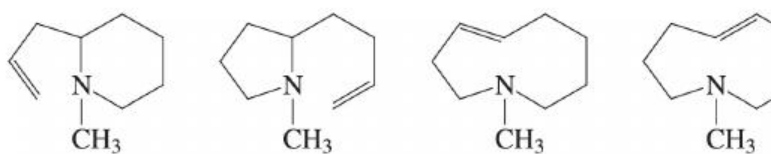
(c) First cycle:



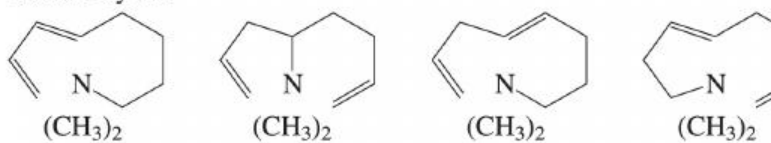
Second cycle:



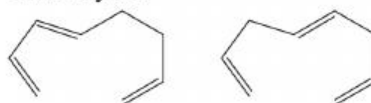
(e) First cycle:



Second cycle:



Third cycle:



16.

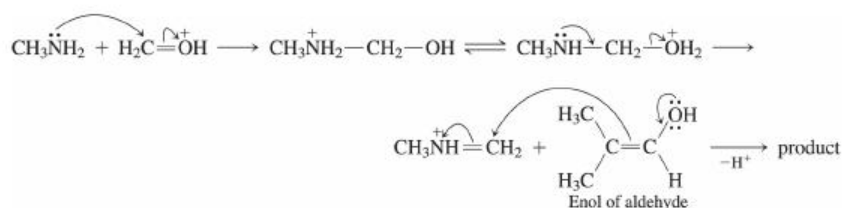
(a) 4-Heptanamine;

(b) 3-heptanamine;

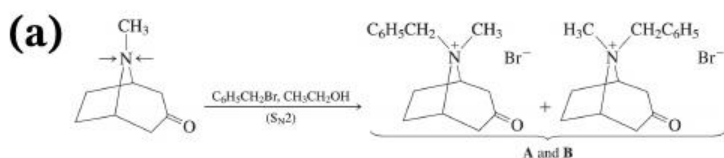
(c) 1-heptanamine;

(d) 2-heptanamine.

17. The reaction takes place under *acidic* conditions.

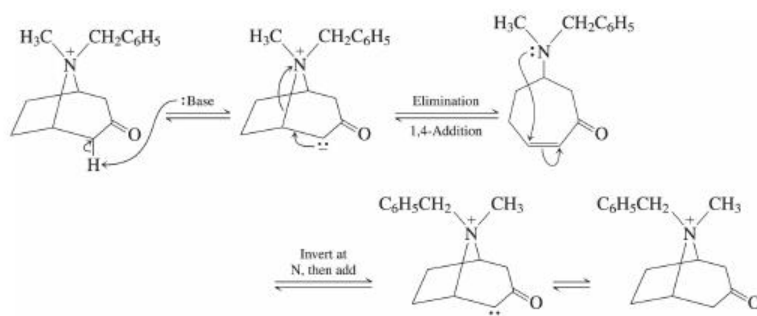


18. Tropinone is a tertiary amine. Alkylation of nitrogen can occur from either the “left” or “right” (arrows, below), giving stereoisomeric products.



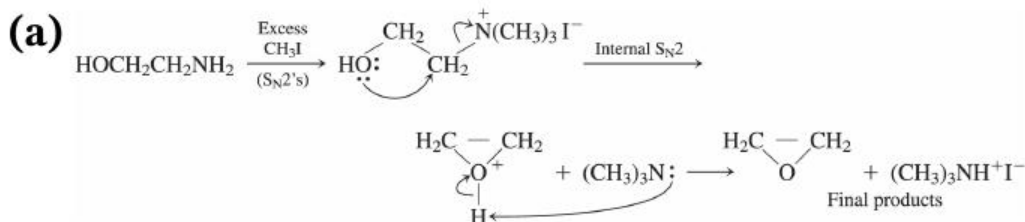
(b) Diastereomers (they are not mirror images)

(c) Where are acidic hydrogens in A and B? At the carbons $\alpha\alpha$ to the ketone carbonyl.

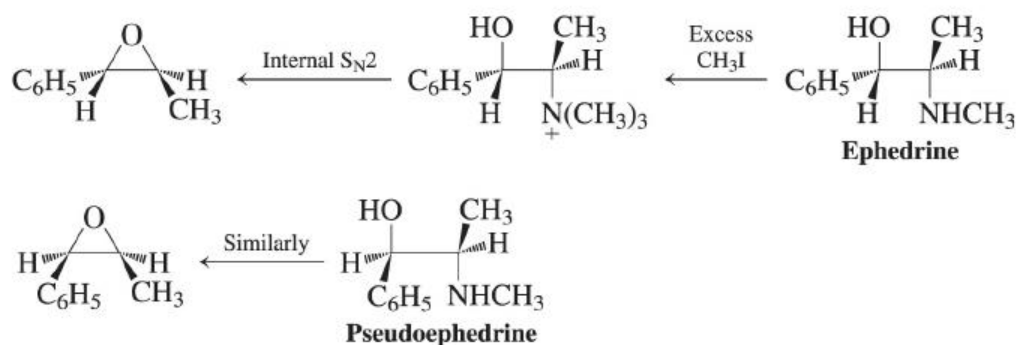


Deprotonation and elimination gives an enone, C.C.
 The amine is free to add back, re-forming the original ketone, or it can invert at nitrogen first and then add back, which gives the stereoisomeric product (CH₃ and C₆H₅CH₂CH₃ and C₆H₅CH₂ groups switched places).

19.

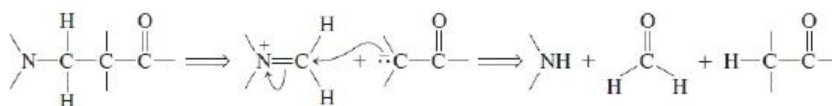


(b) Work backward.

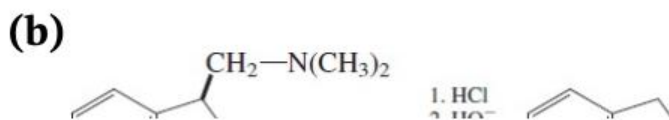
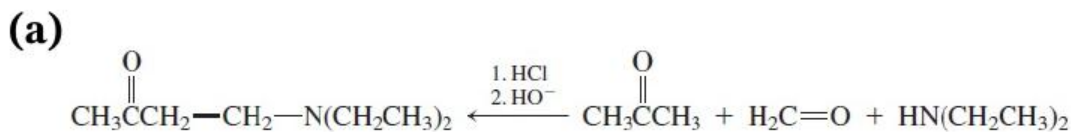


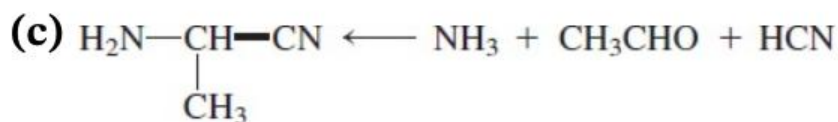
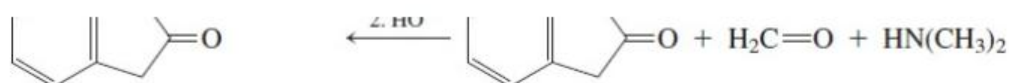
Ephedrine and pseudoephedrine are diastereomers!

50. Analyze the Mannich reaction in terms of the functional unit it constructs.



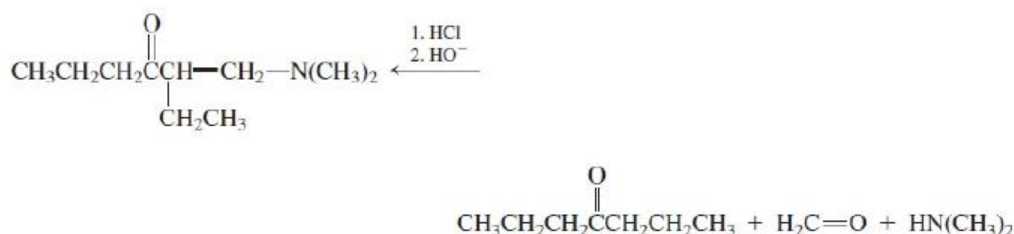
Relevant bonds are emphasized in the answers below.



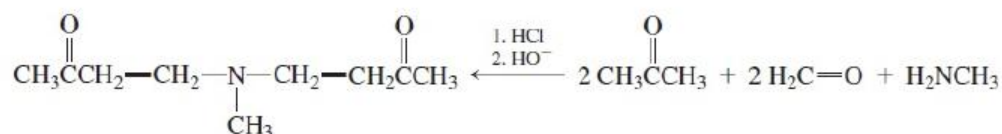


Here the nucleophile is different; it is the cyanide ion that adds to the iminium carbon.

(d)

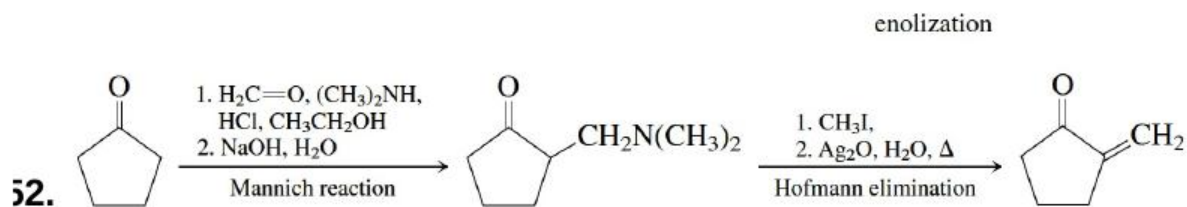
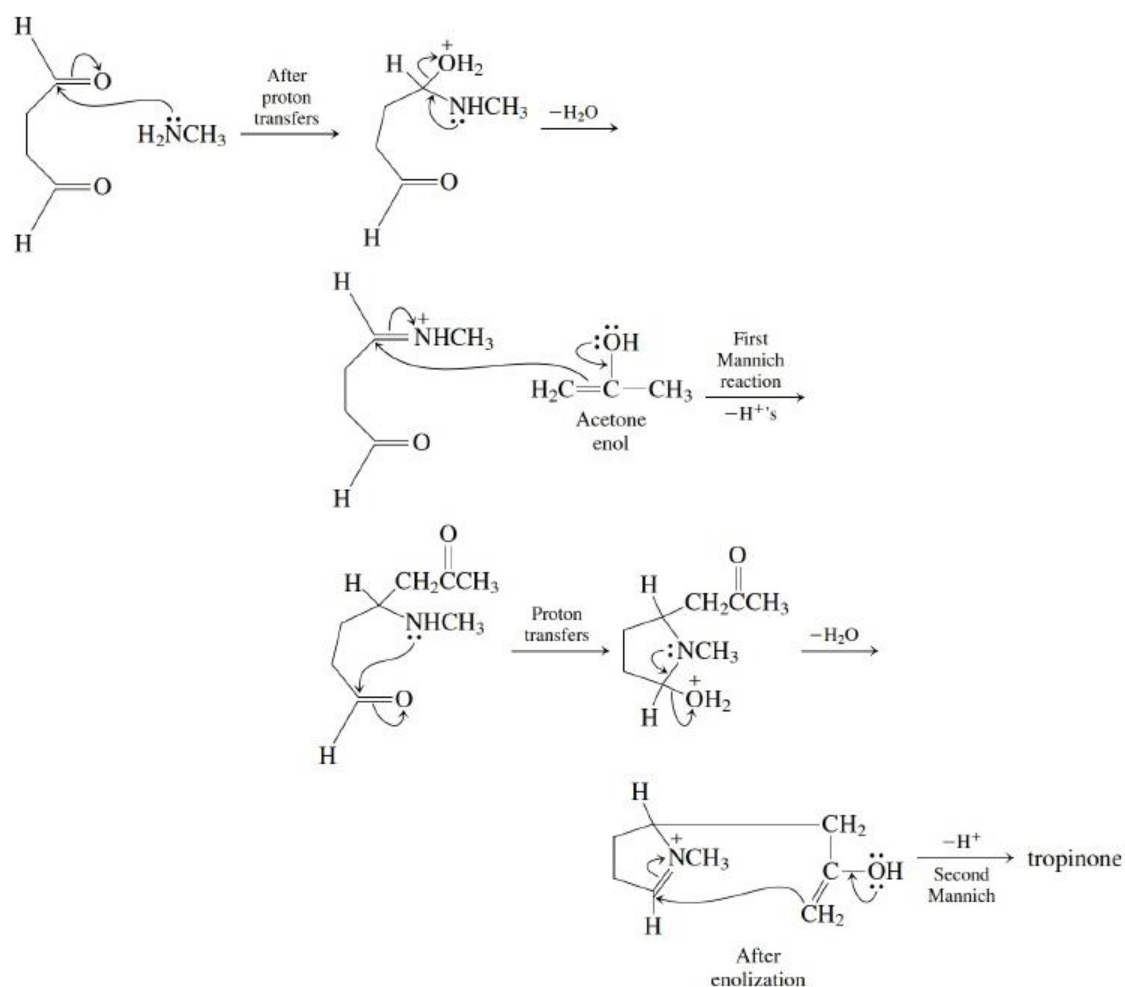


(e)

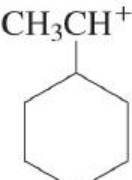


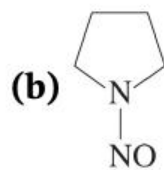
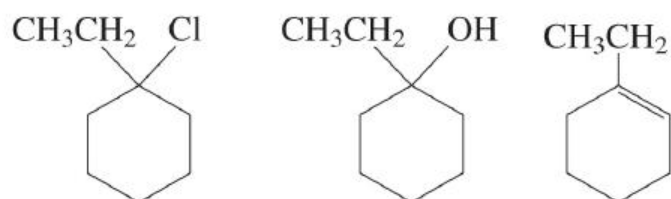
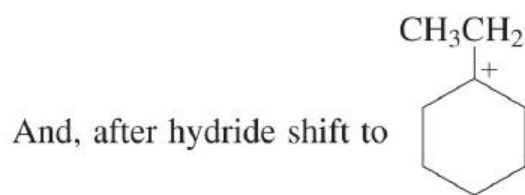
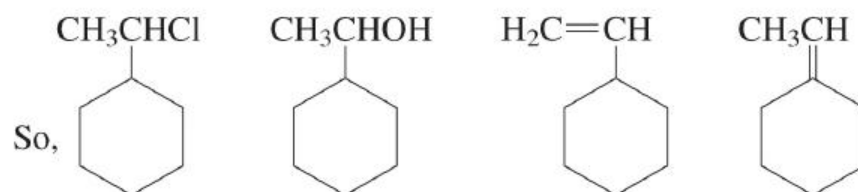
Two Mannich reactions are involved in this example. Note the **primary** amine and the presence of **two** moles each of formaldehyde and acetone.

51. A double Mannich reaction, similar to [Problem 50\(e\)](#):

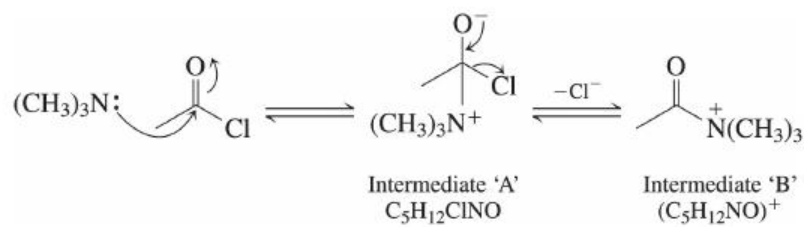


The Mannich reaction–Hofmann elimination sequence is a useful synthesis of α,β - α,β -unsaturated ketones.

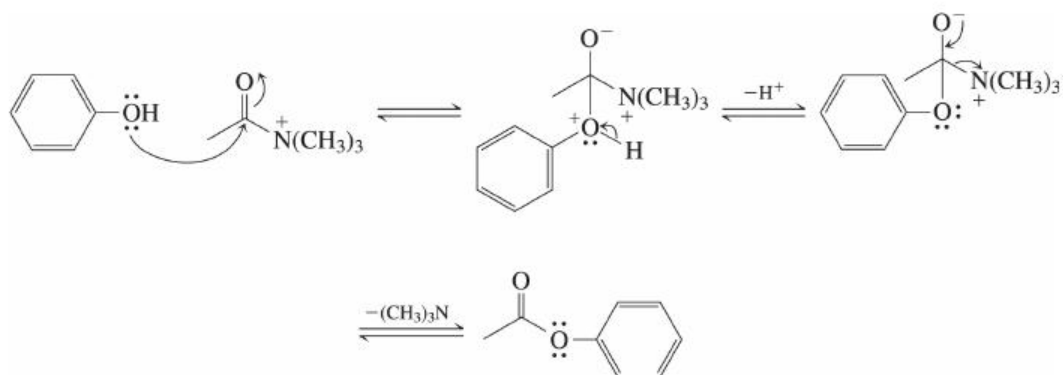
53. (a) All possible products of  !



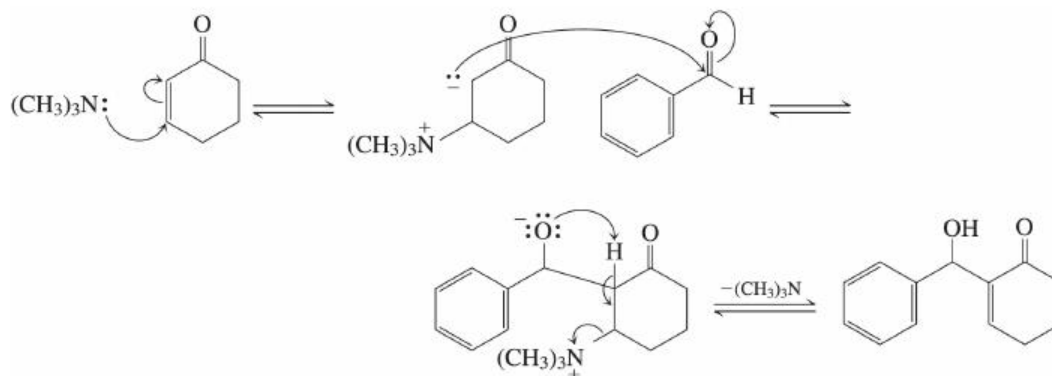
54. (a) Think mechanistically. Nucleophilic nitrogen + reactive carboxylic acid derivative with δ^+ carbonyl carbon atom:



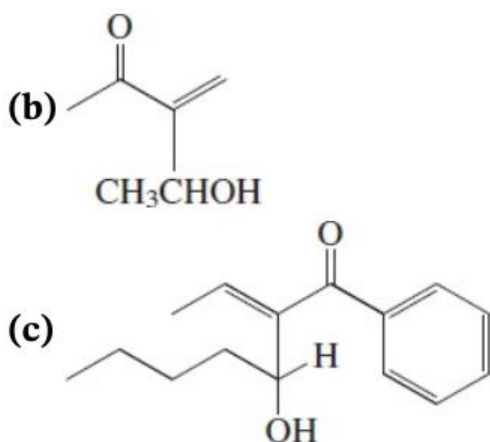
(b)



55. (a) Follow the hint to add the amine. The result is an enolate and, as we saw in [Chapter 17](#), enolates add to aldehydes (aldol reaction). The aldol product has an ammonium ion, a good leaving group, in position to be eliminated to regenerate the α,β double bond of the product:

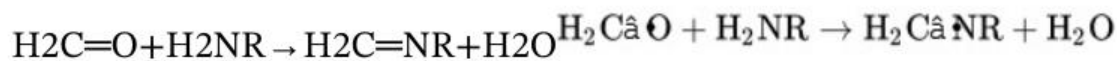


The Baylis-Hillman pattern is connection of the α carbon of the α,β -unsaturated ketone to the carbonyl carbon of the aldehyde, making an alcohol.

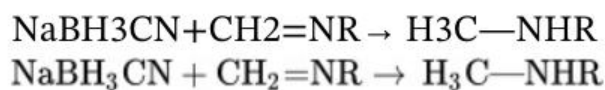


56. Follow the mechanistic pattern **twice**. The first reductive amination forms a secondary amine with one methyl group on nitrogen. A second reductive amination can still take place, in accordance with the pattern shown in Section 21-6, to give the final dimethylated product.

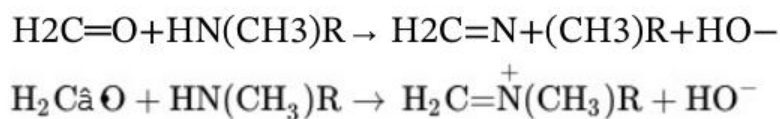
First is (reversible) imine formation:



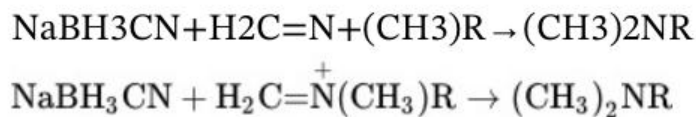
Then reduction gives the secondary amine:



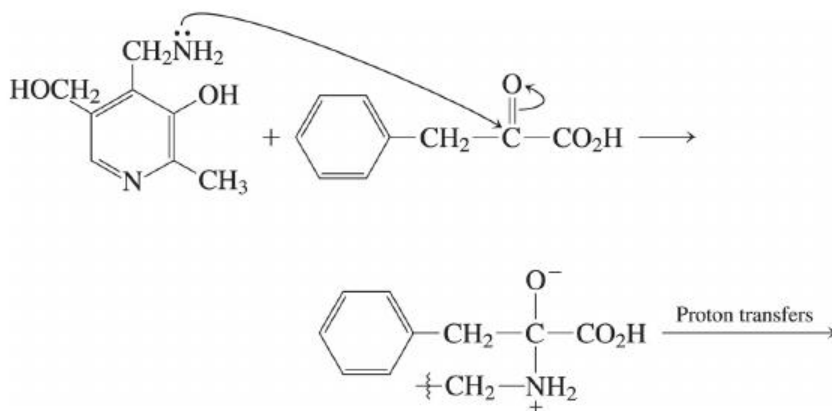
Another $\text{H}_2\text{C}=\text{O}$ reacts, forming an iminium ion:

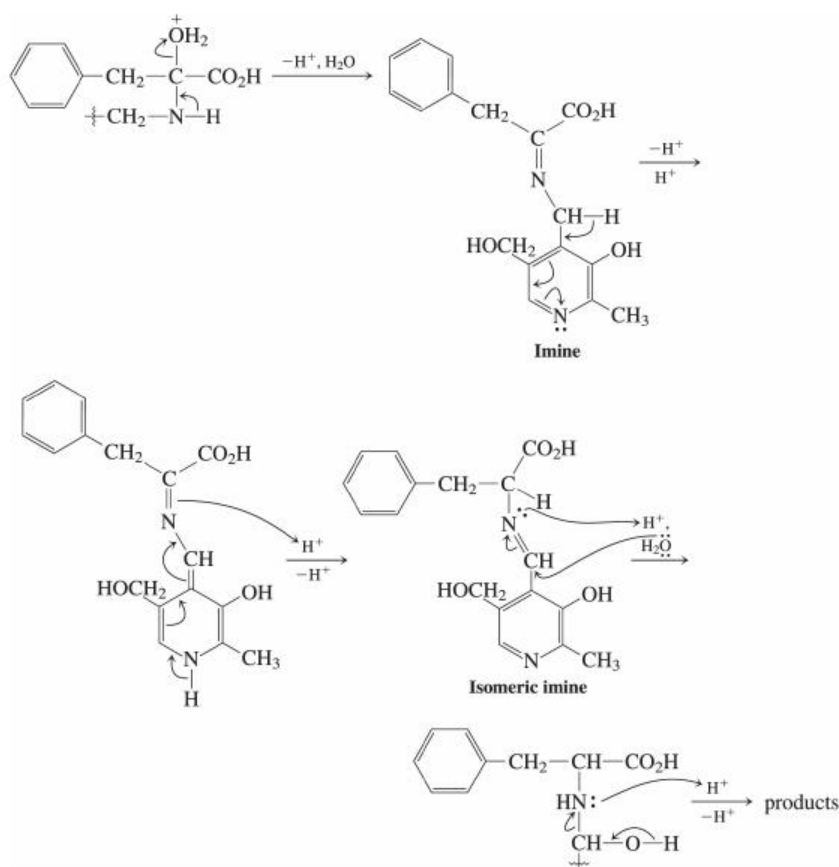


Reduction of this then gives the final product, a dimethylated amine:

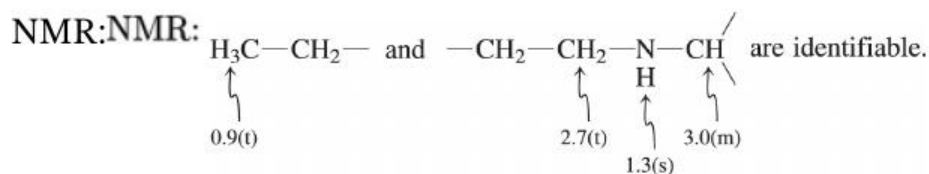


57.





58. IR: Secondary amine.



Total of 17 hydrogens in the molecule.

MS: m/z 127 - 17 (H's) - 14 (N) = 96 or 8 carbons, so $\text{C}_8\text{H}_{17}\text{N}$.

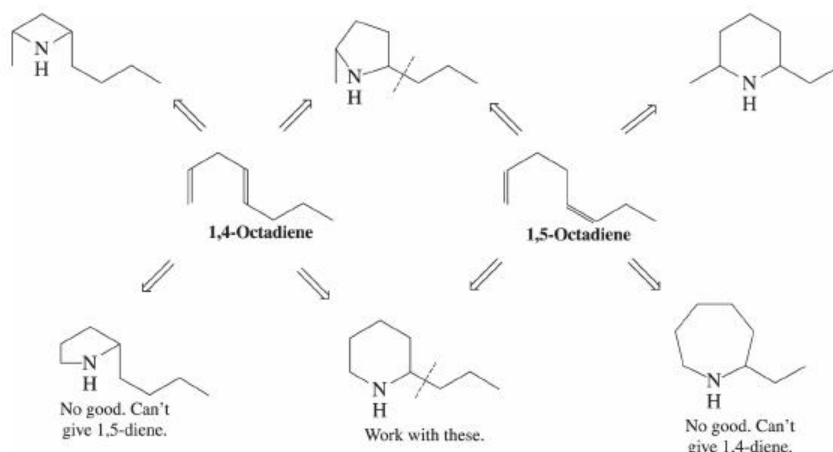
MS: m/z 127 - 17 (H's) - 14 (N) = 96 or 8 carbons, so $\text{C}_8\text{H}_{17}\text{N}$.

$\text{H}_{\text{sat}} = 16 + 2 + 1 = 19$; $\text{H}_{\text{sat}} = 16 + 2 + 1 = 19$; degree of unsaturation =

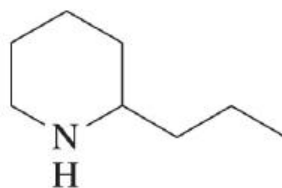
$(19 - 17) / 2 = 1$ degree of unsaturation = $\frac{(19 - 17)}{2} = 1$ π bond or ring present.

MS: Base peak is $[M-43]^+$, $[M-43]^+$, or loss of C_3H_7 , C_3H_7 , perhaps $\underbrace{H_3C-CH_2-CH_2-}_{\text{From NMR}}$.

$H_3C-CH_2-CH_2-$ From NMR. Hofmann elimination results: Reattach NN to alkene carbons in various ways to see what's reasonable. Only structures that can eliminate to give **both** 1,4- and 1,5-octadiene are worth further consideration.

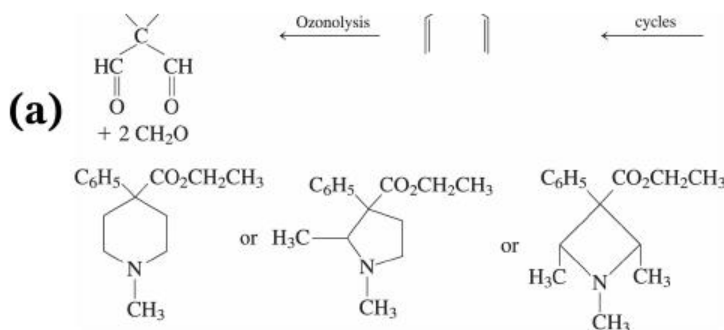


Both structures in the center should lose C_3H_7 , C_3H_7 in the MS (see dashed lines). The top one doesn't fit the NMR though; it should have 2 H's on the carbons attached to N. It also should show a strong $[M-15]^+$, $[M-15]^+$ peak in the MS (loss of CH_3), which is not seen, and it should give another Hofmann product: 2,4-octadiene. The only possible correct structure therefore is



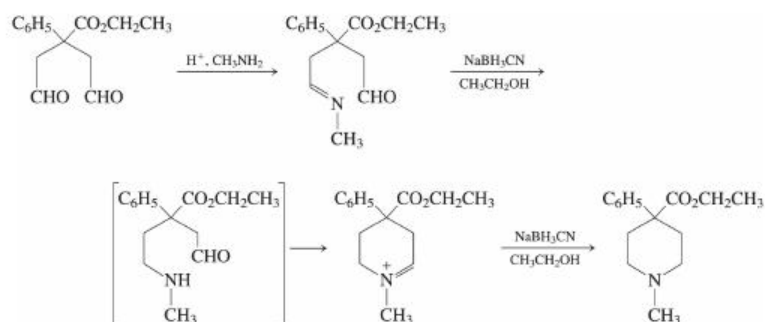
59. Work backward; make sure you include all 15 carbons in your answer.





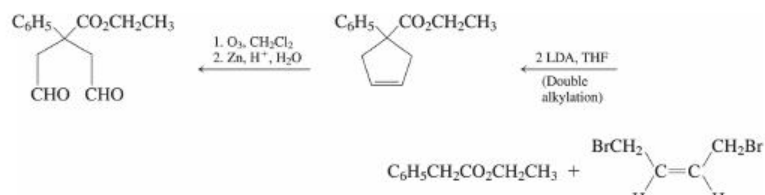
(The extra CH_3 on nitrogen is needed to give the correct molecular formula.) No way yet to choose which of the three is the correct structure.

(b) Big hint: can be converted into pethidine. This strongly suggests that pethidine is the amine with the **six-membered ring** because that one is readily accessible from the dialdehyde as follows:



The sequence is actually carried out all at once, by mixing amine, dialdehyde, and NaBH_3CN together.

The dialdehyde synthesis:



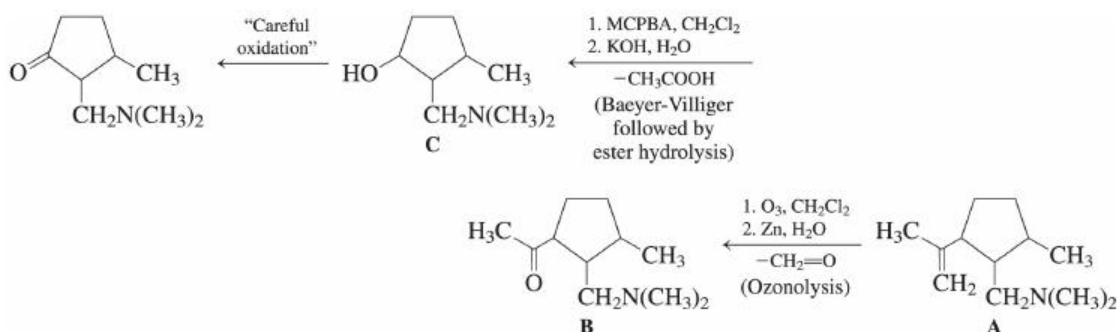
30. $H_{\text{sat}} = 22 + 2 + 1(N) = 25$; degree of unsaturation $= (25 - 21)/2 = 2 \pi$

$$H_{\text{sat}} = 22 + 2 + 1(N) = 25; \text{ degree of unsaturation} = \frac{(25 - 21)}{2} = 2 \pi$$

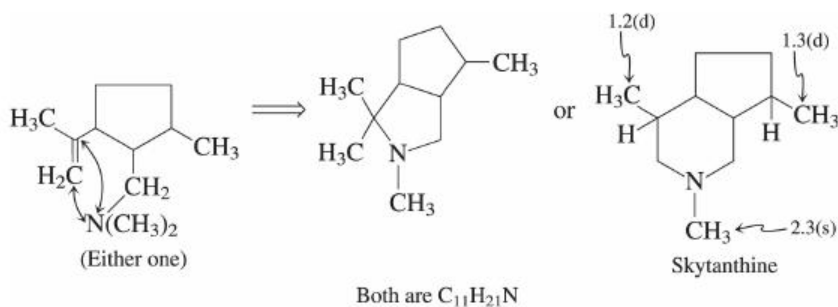
bonds or rings. IR: No $\text{N}-\text{H}$ bonds, so amine is tertiary.

NMR: Two dissimilar $\text{H}_3\text{C}-\text{CH}$ units ($\delta = 1.2$ and 1.3);

($\delta = 1.2$ and 1.3); one unsplit CH_3 (on N, perhaps?) Now work backward.

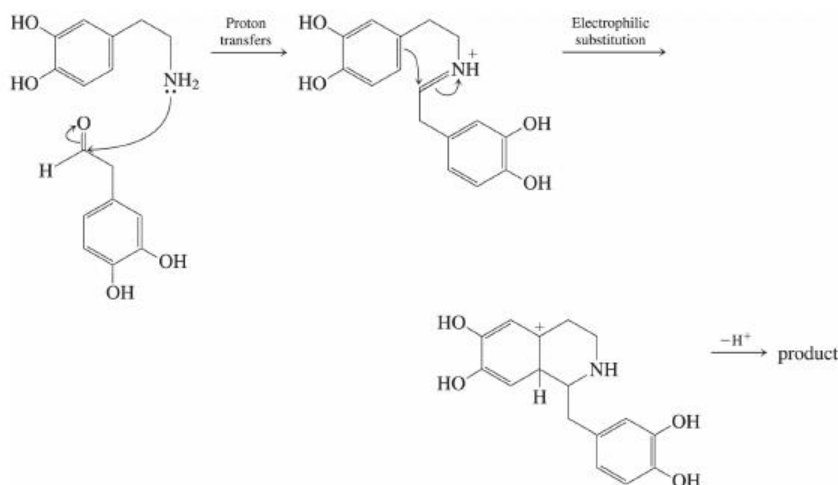


Following everything so far? Now reconnect the nitrogen in A with each of the alkene carbons to establish possible structures before Hofmann elimination.



The three methyl signals in the carbon NMR spectrum match only the second structure, which is the correct one.

31.



The electrophilic substitution is in fact another variation of the Mannich reaction as well, with the electron-rich benzene ring acting as the nucleophile.

32. (a) The catalyst, being a salt, should have some solubility in the water. Because it contains several hydrocarbon substituents on nitrogen, it should also possess some solubility in decane as well.
- (b) The ionic salt NaCNNaCN is essentially insoluble in decane. Thus the concentration of nucleophilic cyanide ion in solution is exceedingly tiny, and SN_2 reaction is correspondingly slow.
- (c) The ammonium cation may exchange counterions (cyanide for chloride) when it is in the water layer, and upon transfer to the decane, cyanide may accompany it. Thus, a significant concentration of cyanide may develop in the decane to participate in SN_2 reaction with the chlorooctane.
33. (c)
34. (c)
35. (e)
36. (a)

37. (d)

22

Chemistry of Benzene Substituents: Alkylbenzenes, Phenols, and Anilines

In this chapter you continue to study the chemistry of aromatic compounds. You find out about reactions at a carbon attached to a benzene ring (a “benzylic” carbon) and about more transformations of benzenes containing either hydroxy or amino groups connected directly to the ring. In all these cases the interaction of the benzene ring with the attached groups modifies their chemistry significantly.

Outline of the Chapter

[22-1 Benzylic Resonance Stabilization](#)

Carbons attached to benzene rings display enhanced reactivity.

[22-2 Benzylic Oxidations and Reductions](#)

Including one especially useful oxidation of alkyl side chains.

[22-3 Naming and Properties of Phenols](#)

[22-4 Preparation of Phenols: Nucleophilic Aromatic Substitution](#)

Useful new methods for synthesizing benzene derivatives.

[22-5 Alcohol Chemistry of Phenols](#)

[22-6 Electrophilic Substitution of Phenols](#)

[22-7 Claisen and Cope Rearrangements](#)

[22-8 Oxidation of Phenols: Benzoquinones](#)

Aromatic alcohols: how the functional groups affect one another.

[22-9 Oxidation – Reduction Processes in Nature](#)

[22-10, 22-11 Arenediazonium Salts and Diazo Coupling](#)

Using aromatic amines to synthesize substituted benzenes.

Keys to the Chapter

22-1. Benzylic Resonance Stabilization

Chemical reactivity at a saturated carbon atom attached to a benzene ring is greatly enhanced over that of ordinary alkyl carbons. Delocalization stabilizes both reaction transition states and intermediates. After reading this section once, go back and take a good look at the sets of resonance forms for the phenylmethyl (benzyl) radical, cation, and anion. There are four forms for each one, leading to increased stability and ease of formation. Most often encountered is nucleophilic substitution on compounds containing potential leaving groups on the benzylic carbon. Both the carbocation intermediate for an $\text{S}_{\text{N}}1$ mechanism and the transition state for the $\text{S}_{\text{N}}2$ mechanism are stabilized, so both pathways may be followed. Just as you saw a long time ago with secondary haloalkanes ([Section 7-5](#)), and more recently with allylic systems ([Sections 14-1](#) through [14-4](#)), the choice of mechanism is determined by the specific conditions and reagents. Also, in analogy with allylic compounds, both free radical and anionic reactions take place at benzylic carbons as well. Notice, however, that the products of these reactions never involve attachment of groups to the benzene ring itself: Double bond migrations, which are often seen with allylic systems, do not occur with benzylic ones, because the aromaticity of the benzene ring would be lost (see [Problem 41](#)).

22-2. Benzylic Oxidations and Reductions

Of the several reactions in this section, the one you are most likely to encounter is the oxidation of alkyl side chains on benzene rings with KMnO_4 to form benzoic acids. Notice two features: First, all carbons but the benzylic carbon are chewed off by the reagent and disappear, and second, an *o,p*-directing alkyl group is converted into

an *m-m*-directing COOHCOOH group.

22-4. Preparation of Phenols: Nucleophilic Aromatic Substitution

The two *nucleophilic* aromatic (*ipso*) substitution reactions are useful synthetic processes. The addition – elimination version is a common reaction of benzenes containing a leaving group together with good electron-withdrawing (anion-stabilizing) groups like NO₂. The elimination – addition (“benzyne”) mechanism occurs when benzenes containing a leaving group, but no other anion-stabilizing groups, are treated with strong bases. Although these nucleophilic substitution reactions are indeed useful, you should keep them in perspective; they are encountered only rarely relative to the electrophilic reactions described earlier. You have already seen the reasons for this: Arenes normally contain electron-rich π systems and are naturally most easily attacked by electrophiles. Only in the presence of strong electron-withdrawing groups or very strong bases will nucleophilic attack be likely to occur.

22-3, 22-5 through 22 -7. Reactions of Phenols

The behavior of aromatic alcohols is so different from that of ordinary alcohols that the properties of the aromatic alcohols merit special coverage. Most of the differences are due to the ability of the benzene ring to delocalize a lone pair of electrons from the phenol oxygen. The immediate result is that phenols are *more acidic* and *less basic* than ordinary alcohols. The conjugate bases of phenols, phenoxide ions, are much less basic than alkoxide ions and can therefore be generated by reaction of phenols with HO–. Being less basic, phenoxide ions are much better leaving groups than are hydroxide or alkoxide ions (see [Problem 62](#) for a practical consequence of this), but they are still good nucleophiles, especially useful in ether synthesis via S_N2 reaction.

The resonance between phenol oxygen and the benzene ring also affects benzene reactivity. The extra electron density in the ring makes it

susceptible to electrophilic attack by even rather weak electrophiles: Reactions with formaldehyde and CO_2 in the presence of base are examples.

22-8. Oxidation of Phenols: Benzoquinones

The reversible redox relationship between 1,4-benzenediols and p-benzoquinones, *p*-benzoquinones, often just called quinones, is a special one because of its relative ease. Conjugate addition and Diels-Alder cycloaddition are common reactions of quinones; conjugate additions have biological importance. The isomeric but less stable 1,2-compounds are encountered much less frequently.

22-9. Oxidation – Reduction Processes in Nature

After a discussion of types of reactions that can damage biological molecules such as lipids in cell membranes, this section illustrates the ways antioxidant molecules such as vitamin E inhibit these processes. These anti-oxidants share one property in common with 1,4-benzenediols: They are easily oxidized, although they usually aren't oxidized to quinones (their structures don't usually permit that). Read carefully to find out about some molecules that are good for you!

22-10 and 22-11. Arenediazonium Salts and Diazo Coupling

Like aromatic alcohols, aromatic amines are also special. In particular, resonance delocalization of the lone pair on nitrogen into the benzene ring makes aromatic amines much less basic than their alkyl relatives. However, most of the reactions of aromatic amines are similar enough qualitatively to those of alkanamines that there's really no need to rehash all that stuff. Instead, this section presents just one class of reactions of these compounds, chosen because of its special versatility in synthesis. Diazotization of a primary aniline produces an arenediazonium salt

containing an —N_2^+ substituent on the benzene ring. The value of these salts lies in the ease of replacement of this group by any of the following (reagents are given in parentheses):

—H (H_3PO_2) —OH ($\text{H}_2\text{O}, \Delta$ or CuOH) —Cl, Br, I (CuX, Δ) —I (HI)
 —CN ($\text{CuCN}, \text{CN}^-, \Delta$)
 —H (H_3PO_2) —OH ($\text{H}_2\text{O}, \Delta$ or CuOH) —Cl, Br, I (CuX, Δ)
 —I (HI) —CN ($\text{CuCN}, \text{CN}^-, \Delta$)

Arenediazonium salts are also electrophilic enough to react with phenols or anilines to form so-called azo dyes via the *diazo coupling* process.